SQL Assignment

ANALYSING THE SIZWE COFFEE BUSINESS

Eight business questions from the Sizwe Coffee year-end review.

COURSE	SQL Fundamentals
DATASET	Sizwe Coffee
SCOPE	DQL - querying only
TASKS	8
DIALECT	PostgreSQL (runs in DuckDB)
SUGGESTED TIME	Approx. 3 hours
ISSUED	October 2026

WITLE PLUS Witle Technologies (Pty) Ltd

The situation

You have joined Sizwe Coffee as a junior data analyst. The financial year ended on 30 June 2025 and the year-end review is in two weeks.

Nomsa Dlamini, the Financial Director, and **Bongani Twala**, the National Sales Manager, have each sent you a list of things they want to understand before that meeting.

The dataset

Everything in this exercise runs on the **Sizwe Coffee** dataset.

Sizwe Coffee (Pty) Ltd is a coffee roastery and wholesale supplier based in Isando, Johannesburg, with a second distribution centre in Epping, Cape Town. It roasts green beans and sells to about 1,400 business customers - grocery chains, restaurant groups, independent cafés, hotels and offices. It does not own retail stores.

Revenue	R441m in FY2024, R480m in FY2025
Staff	320
Customers	1,400
Channels	Wholesale (70% of revenue) and Subscription (30%)
Financial year	1 July to 30 June
Period in the data	1 July 2023 to 30 June 2025

Tables you will use

Table	Rows	What it holds
`customers`	1,400	One row per customer - name, channel, segment, province, credit limit, assigned sales representative
`products`	23	One row per SKU - product line, pack size, list price, unit cost
`sales_orders`	48,144	One row per order - customer, order date, delivery date, channel, sales representative
`sales_lines`	168,172	One row per order line - SKU, units, unit price, discount, line revenue, cost of goods
`ar_ledger`	48,144	One row per customer invoice - amount, due date, amount settled, settlement date
`reps`	18	Sales representatives - territory, monthly target
`rep_visits`	9,400	Visits made by sales representatives to customers, with the outcome
`trade_promotions`	640	Promotional support given to a customer on a SKU
`vendors`	74	Suppliers - type, payment terms

How the tables connect

- `sales_orders.customer_id` → `customers.customer_id`
- `sales_orders.order_id` → `sales_lines.order_id`
- `sales_lines.sku` → `products.sku`
- `sales_orders.rep_id` and `customers.rep_id` → `reps.rep_id`
- `ar_ledger.customer_id` → `customers.customer_id`
- `rep_visits.customer_id` and `trade_promotions.customer_id` → `customers.customer_id`

Two things to notice before you start. **Revenue lives on `sales_lines`, not on `sales_orders`** - an order has no value of its own, only the sum of its lines. And **`channel` appears on both `customers` and `sales_orders`** - use whichever the question points at, and be ready to explain why they agree.

What to hand in

Submit a link to a public GitHub repository containing your work. The repository must hold two files:

- `queries.sql` - all eight queries, in order, each one preceded by a comment giving the task number.
- `README.md` - a well-documented write-up of the project (see below).

Where a task asks for a written answer, put it in a comment directly below that query. Two or three sentences is enough - you are being marked on whether you can read your own output, not on length.

```
-- Task 1: Channel performance
SELECT ...

-- Answer: <your reading of the result goes here, in two or three sentences>
```

Your README should read as a standalone project write-up that someone who has never seen this brief can follow. Include:

- **Project overview:** what the project is, the Sizwe Coffee business context and the eight questions you were asked to answer.
- **Dataset:** the tables you used and how they connect.
- **Skills demonstrated:** the SQL techniques you applied (for example joins, aggregation, CASE expressions, CTEs and window functions) and the task where each one was used.
- **How to run:** the database or tool you used and the steps needed to reproduce your results.
- **Key insights:** your main findings for Nomsa and Bongani, each backed by specific numbers from your results.

The tasks

Task 1 - Channel performance

Nomsa needs the revenue split by channel for FY2024 and FY2025 side by side, with the growth rate between them. One row per channel, three figures on each.

FY2024 is orders placed on or before 30 June 2024. FY2025 is everything after that.

Task 2 - Where the profit actually comes from

For each product line, return revenue, cost of goods, gross profit and gross margin percentage. Sort by gross profit.

In one or two sentences under your query: which line earns the most profit, and is it the same line that earns the highest margin?

Task 3 - The collections problem

Nomsa suspects the business is being paid more slowly than it used to be, and wants to know where the problem sits.

For each channel, return the number of invoices, how many have been settled, how many are still open, the average number of days taken to settle, and the total value still outstanding.

In two or three sentences: what does this tell you, and which channel would you investigate first?

Task 4 - The ten accounts that matter most

Bongani wants a one-page list of the ten largest customers by **FY2025** revenue, showing the trading name, segment, province, the sales representative who owns the account, and the revenue.

Not every customer has a sales representative. Make sure your query does not silently drop anyone.

Task 5 - A commercial gap

Trade promotions are how Sizwe supports its wholesale accounts. Bongani thinks some sizeable customers have been getting nothing.

Return the fifteen highest-revenue **Wholesale** customers who have **never** received a trade promotion, with their segment and province.

One sentence: what would you recommend doing about the top three?

Task 6 - Are the sales representatives hitting target?

Each sales representative carries a monthly revenue target. Return every sales representative with their territory, their monthly target, their actual FY2025 monthly average, and that average as a percentage of target. Sort by percentage of target, best first.

Every sales representative must appear, including any who generated no revenue.

Task 7 - The shape of the year

Return monthly revenue across the full two years, with the percentage change on the previous month and a rolling three-month average alongside it.

In two sentences: is there a seasonal pattern, and when is Sizwe's quietest month?

Task 8 - Why has the margin moved?

Gross margin fell from 34.7% in FY2024 to 33.0% in FY2025, while revenue grew nearly 9%. Nobody can explain it.

For each product line, return revenue and gross margin for each financial year, plus that line's **share of total revenue** in that year.

Then answer, in a short paragraph: what caused the margin decline? Your answer must point at specific numbers in your result.